

3. A food colouring contains a mixture of yellow and green dyes.
The table shows the R_f values of these dyes using different solvents in separate chromatography experiments.

Colour	R_f value when the solvent is		
	Water	Acid	Alcohol
yellow	0.74	0.35	0.00
green	0.76	0.81	0.46

(a) Why would yellow and green dyes be difficult to separate in a chromatography experiment using **water** as the solvent? [1]

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(b) Describe what you would expect to see in the chromatogram obtained using **alcohol** as the solvent. [1]

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- (c) Complete the chromatogram to show the results you would expect using **acid** as the solvent. [2]

